



Department of Computer Science & Engineering (Cyber Security)

**Bachelor of Engineering – 7th Semester
Major Project Phase – 1 Synopsis**

TrustEdge : TPM-Assisted Secure Attestation Framework for AI Agents

Name: Manyu Simha Ravi

Name: Letitia Gilbert

Name: Surendra S

USN: 1DS23CY035

USN: 1DS23CY031

USN: 1DS23CY059

Guide Name & Designation : Dr. Pallavi Bindagi, Assistant Professor (Dept Of CSCY)

Introduction:

Artificial Intelligence (AI) agents are increasingly being deployed to automate tasks such as file access, database queries, cloud interactions, and tool execution. While these agents improve productivity, they also introduce significant security concerns because they often operate with elevated privileges and access to sensitive resources. Existing authentication mechanisms verify the identity of an agent during deployment but do not continuously verify its integrity during execution. This project proposes TrustEdge, a hardware-assisted runtime attestation framework that verifies the trustworthiness of an AI agent before allowing privileged operations, thereby improving the security of autonomous AI systems.

Problem Statement:

Current AI agent deployments rely primarily on authentication and authorization mechanisms, which do not guarantee that the execution environment remains trustworthy after deployment. Runtime modifications, policy tampering, compromised tools, or malicious models can cause an AI agent to perform unauthorized actions while appearing legitimate. There is a need for a lightweight framework that continuously verifies the integrity of AI agents before sensitive operations using hardware-assisted attestation.

Objectives :

- Design a hardware-assisted runtime attestation framework for AI agents.
- Develop a policy-driven mechanism to verify privileged operations before execution.
- Implement a prototype using Docker, Software TPM (swtpm), and a locally deployed language model to evaluate the proposed framework.

Literature Review:

Sl. No.	Paper Title	Year	Journal / Conference	Key Findings
1	TrustVisor: Efficient TCB Reduction and Attestation	2010	IEEE S&P	Lightweight hypervisor with TPM attestation.
2	Intel SGX Explained	2016	IACR	Explains Intel SGX architecture and security.
3	C-FLAT: Control-Flow Attestation for Embedded Systems Software	2016	IEEE	Detects runtime control-flow attacks.
4	SCONE: Secure Linux Containers with Intel SGX	2016	OSDI	Protects containerized applications using SGX.
5	Sanctum: Minimal Hardware Extensions for Strong Software Isolation	2016	USENIX Security	Provides secure hardware isolation.
6	Inferring Fine-Grained Control Flow Inside SGX Enclaves with Branch Shadowing	2017	USENIX Security	Demonstrates SGX side-channel attacks.
7	Gramine: A Library OS for Confidential Computing	2020	EuroSys	Enables secure execution inside enclaves.
8	Keystone: An Open Framework for Architecting Trusted Execution Environments	2020	EuroSys	Open-source TEE framework for RISC-V.
9	Trusted Platform Module (TPM) 2.0 Library Specification	2023	TCG	Defines TPM 2.0 architecture and commands.
10	Runtime Integrity Verification Using TPM	2023	IEEE	TPM-based runtime integrity verification.
11	Trusted Execution Environments: A Survey	2023	ACM CSUR	Comprehensive survey of TEE technologies.
12	Profiling with Trust: System Monitoring from Trusted Execution Environments	2023	USENIX	Trusted monitoring using TEEs.

13	Scalable Runtime Remote Attestation for Complex Systems	2023	IEEE	Scalable runtime attestation framework.
14	Universal Remote Attestation for Cloud and Edge Platforms	2023	ARES 2023	Proposes a hardware-agnostic remote attestation framework for cloud and edge platforms.
15	A Survey of Remote Attestation	2024	IEEE	Reviews modern remote attestation techniques.
16	Confidential Computing Consortium Architecture	2024	CCC	Overview of confidential computing platforms.
17	Hardware Roots of Trust for Secure Systems	2024	ACM	Reviews hardware root-of-trust mechanisms.
18	WATZ: A Trusted WebAssembly Runtime Environment with Remote Attestation for TrustZone	2024	IEEE	Secure WebAssembly runtime using TrustZone.
19	CCxTrust: Confidential Computing Platform Based on TEE and TPM Collaborative Trust	2024	IEEE	Combines TPM and TEE for trusted computing.
20	Separate but Together: Integrating Remote Attestation into TLS	2024	IEEE	Integrates attestation with secure TLS sessions.
21	Hardware-Assisted Security for AI Agents	2025	IEEE	Hardware-backed trust for AI agent security.
22	Agent Security Bench	2025	arXiv	Benchmarks security risks in AI agents.
23	AI Supply Chain Security: Threats and Defenses	2025	IEEE	Surveys AI model supply-chain attacks.
24	TEEcorrelate: An Information-Preserving Defense against Performance-Counter Attacks on TEEs	2025	IEEE	Defends against TEE side-channel attacks.
25	A Survey on	2025	IEEE	Surveys

	Confidential Computing			confidential computing technologies.
--	------------------------	--	--	--------------------------------------

Methodology, Tools & Techniques:

The proposed framework will be implemented using two Docker containers: one hosting the TrustEdge framework and the AI agent, and the other hosting a Software Trusted Platform Module (swtpm). The framework consists of a Policy Engine, Integrity Monitor, Attestation Manager, and Verifier. SHA-256 hashing will be used to generate integrity measurements, while the TPM will provide hardware-backed cryptographic attestation. The AI agent will be deployed locally using Ollama and a lightweight language model. Sensitive operations will be evaluated through policy checks and runtime attestation before execution.

Tools and Technologies

- Python
- Docker & Docker Compose
- Ollama
- Qwen/Llama 3B
- Software TPM (swtpm)
- TPM2-Tools / TPM2-TSS
- SHA-256

Expected Outcomes:

- A functional prototype of the TrustEdge framework demonstrating hardware-assisted runtime attestation.
- Detection and prevention of unauthorized or modified AI agent execution before privileged actions.
- Evaluation of the framework using normal and attack scenarios to demonstrate its effectiveness and feasibility.

References:

[1] J. M. McCune, Y. Li, N. Qu, Z. Zhou, A. Datta, V. Gligor, and A. Perrig, "TrustVisor: Efficient TCB Reduction and Attestation," *Proceedings of the IEEE Symposium on Security and Privacy (S&P)*, 2010.

[2] V. Costan and S. Devadas, "Intel SGX Explained," *IACR Cryptology ePrint Archive*, 2016.

[3] T. Abera *et al.*, "C-FLAT: Control-Flow Attestation for Embedded Systems Software," *IEEE*, 2016.

[4] S. Arnaudov *et al.*, "SCONE: Secure Linux Containers with Intel SGX," *Proceedings of the 12th USENIX Symposium on Operating Systems Design and Implementation (OSDI)*, 2016.

[5] V. Costan, I. Lebedev, and S. Devadas, "Sanctum: Minimal Hardware Extensions for Strong Software Isolation," *Proceedings of the 25th USENIX Security Symposium*, 2016.

[6] M. Lee, T. Kim, and others, "Inferring Fine-Grained Control Flow Inside SGX Enclaves with Branch Shadowing," *Proceedings of the 26th USENIX Security Symposium*, 2017.

[7] A. Tsai *et al.*, "Gramine: A Library OS for Confidential Computing," *Proceedings of EuroSys*, 2020.

[8] D. Lee *et al.*, "Keystone: An Open Framework for Architecting Trusted Execution Environments," *Proceedings of the Fifteenth European Conference on Computer Systems (EuroSys)*, 2020.

[9] Trusted Computing Group, "Trusted Platform Module Library Specification, Family 2.0, Level 00, Revision 1.59," Trusted Computing Group, 2023.

[10] *Runtime Integrity Verification Using TPM*, IEEE, 2023.

[11] "Trusted Execution Environments: A Survey," *ACM Computing Surveys*, 2023.

[12] "Profiling with Trust: System Monitoring from Trusted Execution Environments," *USENIX*, 2023.

[13] "Scalable Runtime Remote Attestation for Complex Systems," *IEEE*, 2023.

- [14] S. Ott, M. Kamhuber, J. Pecholt, and S. Wessel, "Universal Remote Attestation for Cloud and Edge Platforms," *Proceedings of the 18th International Conference on Availability, Reliability and Security (ARES)*, 2023.
- [15] M. Ambrosin, M. Conti, A. Ibrahim, G. Neven, and M. Sadeghi-Yekta, "A Survey on Remote Attestation," *IEEE Communications Surveys & Tutorials*, 2024.
- [16] Confidential Computing Consortium, "Confidential Computing Architecture," Linux Foundation, 2024.
- [17] "Hardware Roots of Trust for Secure Systems," *ACM*, 2024.
- [18] Y. Xu *et al.*, "WATZ: A Trusted WebAssembly Runtime Environment with Remote Attestation for TrustZone," *IEEE*, 2024.
- [19] K. Shang, J. Lin, Y. Qin, M. Shen, H. Ma, W. Feng, and D. Feng, "CCxTrust: Confidential Computing Platform Based on TEE and TPM Collaborative Trust," 2024.
- [20] "Separate but Together: Integrating Remote Attestation into TLS," *IEEE*, 2024.
- [21] "Hardware-Assisted Security for AI Agents," *IEEE*, 2025.
- [22] "Agent Security Bench," *arXiv*, 2025.
- [23] "AI Supply Chain Security: Threats and Defenses," *IEEE*, 2025.
- [24] "TEEcorrelate: An Information-Preserving Defense against Performance-Counter Attacks on TEEs," *IEEE*, 2025.
- [25] "A Survey on Confidential Computing," *IEEE*, 2025.

Date: 13/08/2026

Signature of Student(s):

Signature of Guide: